Construct Voronoi diagrams

1 Construct a Voronoi diagram for the following sites:

a $A(-4,1)$, $B(0,-3)$

b $A(-2,-4)$, $B(7,2)$

c $A(-4,2)$, $B(6,-4)$, $C(0,4)$

d $A(-5,-1)$, $B(3,3)$, $C(5,-7)$

2 Consider the sites $X(-4,2)$ and $Y(2,-6)$.

a Construct a Voronoi diagram for these sites.

b Which site is the point $(-1,-1)$ closest to?

c State the coordinates of a point that is equidistant from both sites.

3 Consider the sites $X(-3,2), Y(1,-6)$ and $Z(0,4)$.

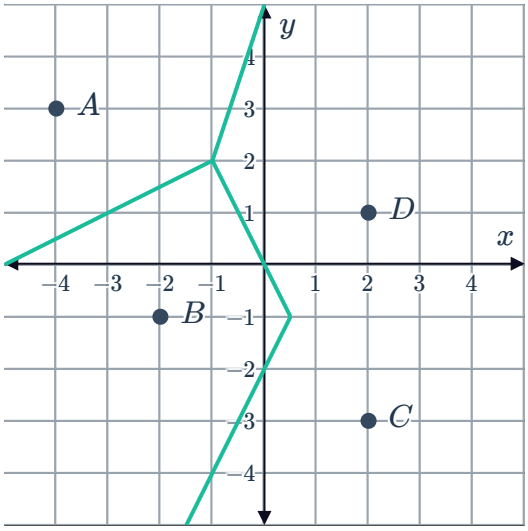
a Construct a Voronoi diagram for these sites.

b Which site(s) is the point $(1,3)$ closest to?

4 Consider the following Voronoi diagram:

a Explain how we know that the Voronoi diagram has an edge missing.

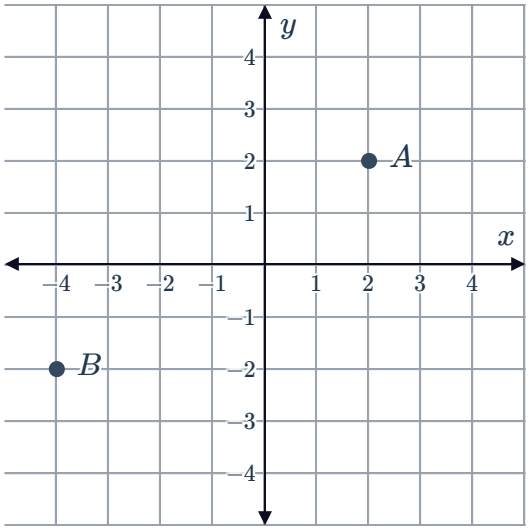
b Complete the diagram by adding the missing edge.



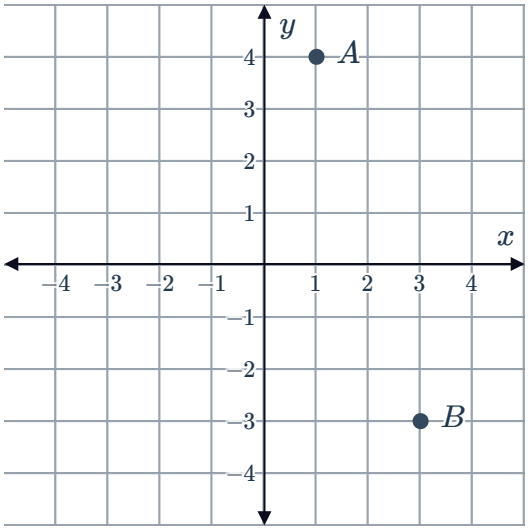
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5 Construct a Voronoi diagram for the following sites:

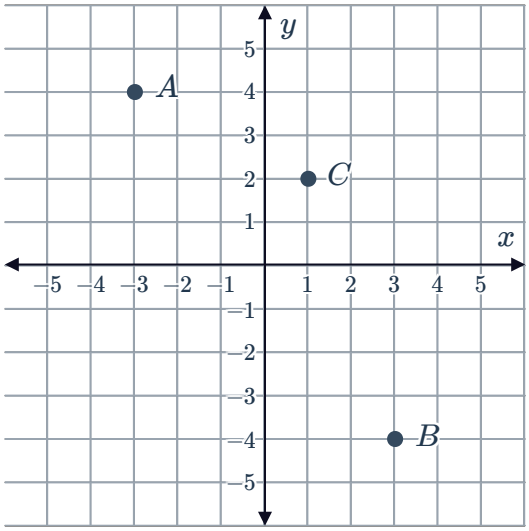
a



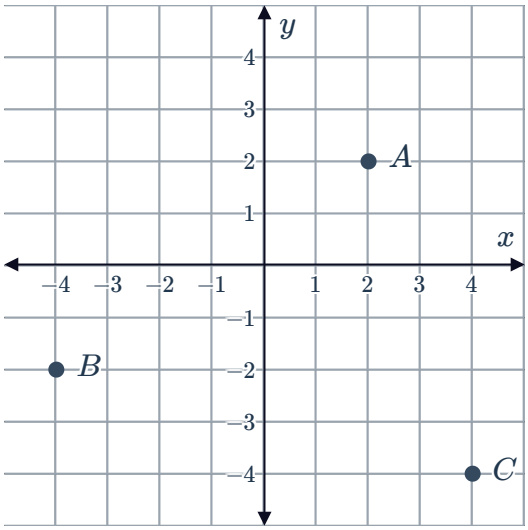
b



c



d



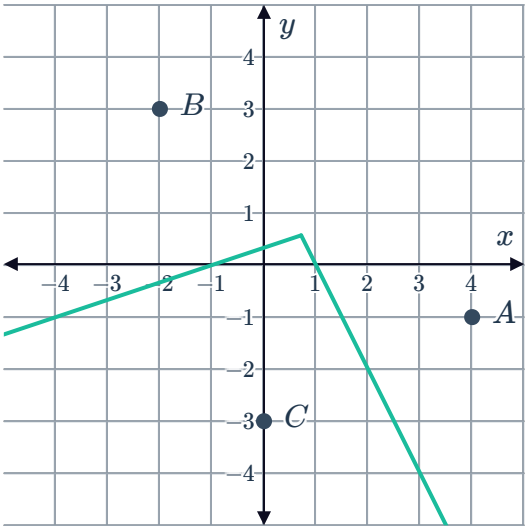
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6 For each of the following Voronoi diagrams:

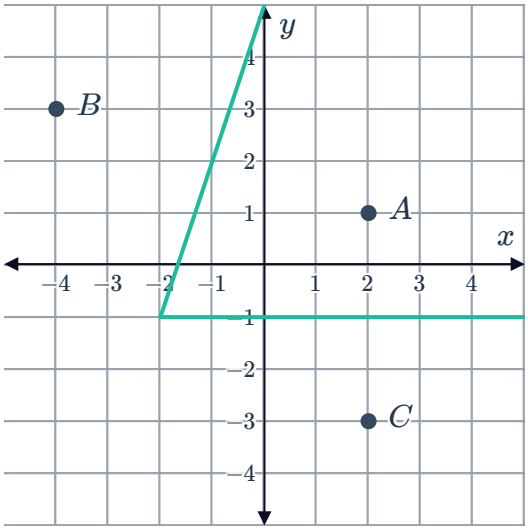
i Find the equation of the missing edge.

ii Complete the Voronoi diagram.

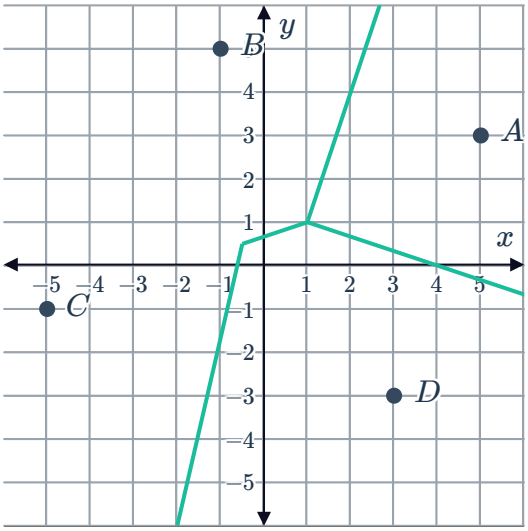
a



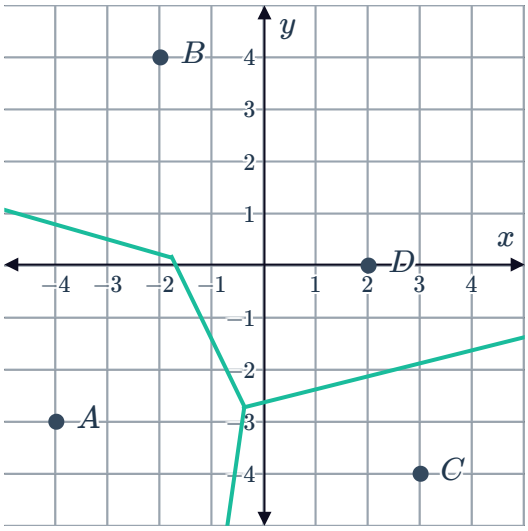
b



c



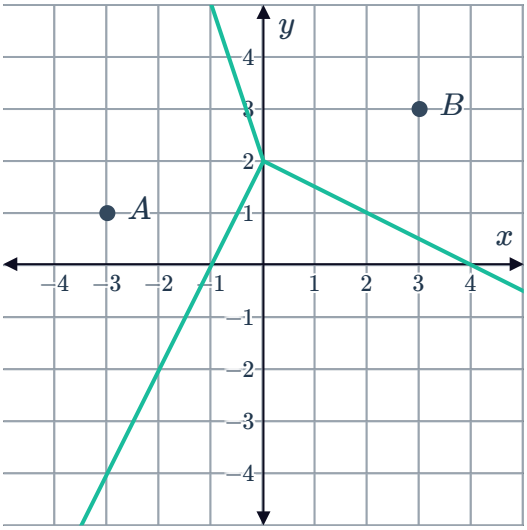
d



7 Consider the following Voronoi diagram:

a Explain how we know that the Voronoi diagram has a missing site.

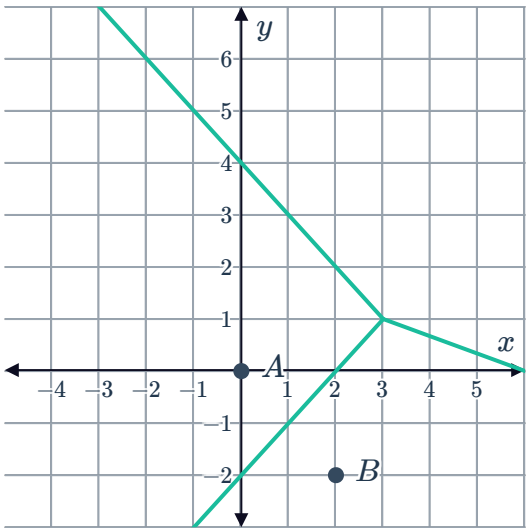
b Find the coordinates of the missing site.



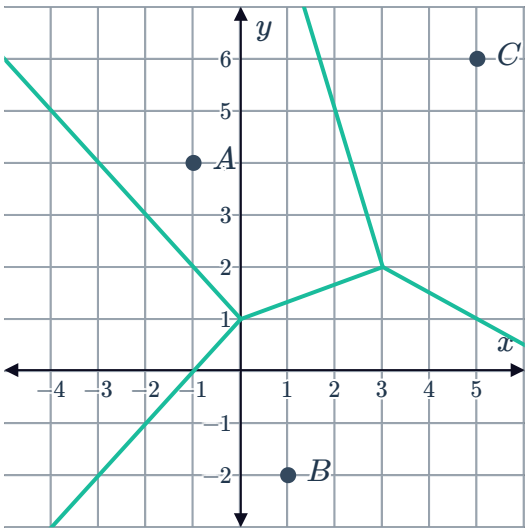
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8 Find the coordinates of the missing site in each of the following Voronoi diagrams:

a

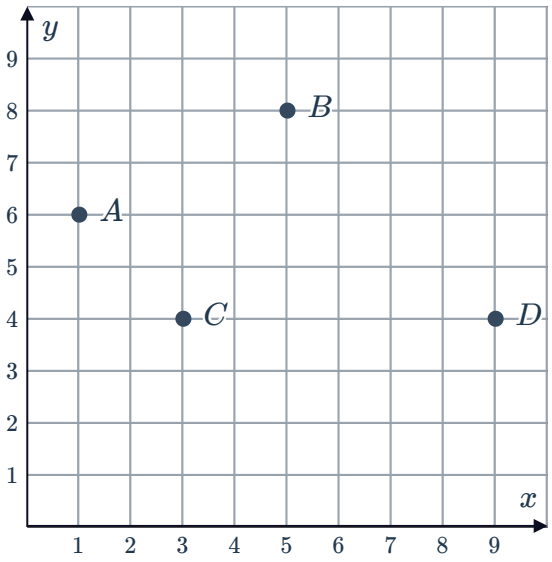


b



Applications

9 The following map shows the locations of the local bus stops in a suburb. The units are in kilometres.

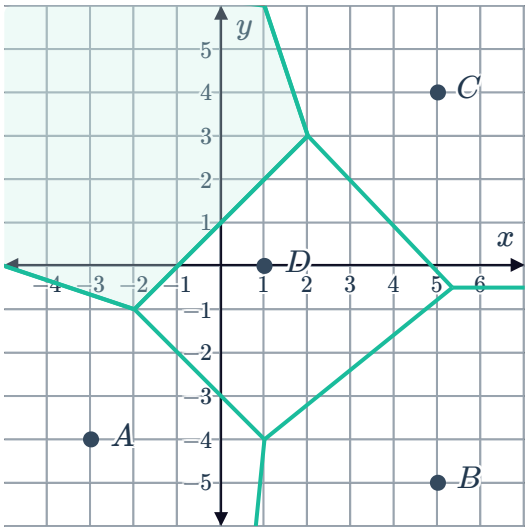


a Construct a Voronoi diagram for the sites of the bus stops.

b If Juan lives at $(5, 4)$, which bus stop is closest to him?

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10 The following map shows the locations of light rail stations in a city. The five regions are suburbs, and only four of the five suburbs have a light rail station. The units are in kilometres.
The people that live in the shaded suburb want a light rail station built in their suburb. But the people that live on the edges of that suburb want to be able to use either station in their adjacent suburbs.
Where should the light rail station be built in the shaded suburb, so that the people living on the edges are equally distant from both light rails in their adjacent suburbs?



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